A FUNCTOR WITH AN EXACT LEFT ADJOINT PRESERVES INJECTIVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all the implications, and each of them is written out inside the counted span. The statement opens with the author's own slogan environment, a comment-type environment of the chapter preamble, which is delivered as part of the counted statement.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Exactness properties of adjoint functors. Categories, Lemma categories-lemma-exact-adjoint. Supplies that the left adjoint v is right exact, which together with the characterization of exactness gives the equivalence of (a) and (b), and is used again inside the two implications.
- Characterizations of exact functors. Same chapter, Lemma lemma-exact-functor. Supplies the characterization of one-sided exactness through short exact sequences that turns injectivity-preservation of v into exactness of v.
- Injective objects and the extension property. Same chapter, the definition of an injective object. Supplies the extension property used twice: once with I to extend a map out of vN, and once with u(I) to extend a map out of B.
- Categories with enough injectives. Same chapter, the definition of a category with enough injectives. Supplies the monomorphism A -> I out of the kernel used in the last part of the proof.

Source reference. Original lemma 0039: sources/stacks-categories/pinned-original-source.tex, source lines 3727–3727. Navigation pointer invoked at proof line 7506; the counted span contains the pointer only and the cited result is not reproduced

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Source reference. Original lemma 010N: sources/stacks-homology/pinned-original-source.tex, source lines 1762–1762. Navigation pointer invoked at proof line 7507; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $u : \mathcal{A} \to \mathcal{B}$ and $v : \mathcal{B} \to \mathcal{A}$ be additive functors with u right adjoint to v. Consider the following conditions:*

(a) *v transforms injective maps into injective maps,*
(b) *v is exact, and*
(c) *u transforms injectives into injectives.*

Then (a) $\Leftrightarrow$ (b) $\Rightarrow$ (c). If $\mathcal{A}$ has enough injectives, then all three conditions are equivalent.

Proof. Observe that v is right exact as a left adjoint (Categories, Lemma 0039). Combined with Lemma 010N this explains why (a) $\Leftrightarrow$ (b).

Assume (a). Let I be an injective object of $\mathcal{A}$. Let $\varphi : N \to M$ be an injective map in $\mathcal{B}$ and let $\alpha : N \to uI$ be a morphism. By adjointness we get a morphism $\alpha : vN \to I$ and by assumption $v\varphi : vN \to vM$ is injective. Hence as I is an injective object we get a morphism $\beta : vM \to I$ extending α. By adjointness again this corresponds to a morphism $\beta : M \to uI$ extending α. Hence (c) is true.

Assume $\mathcal{A}$ has enough injectives and (c) holds. Let $f : B \to B'$ be a monomorphism in $\mathcal{B}$, and let $A = \mathrm{Ker}(v(f))$. Choose a monomorphism $g : A \to I$ with I injective (it exists by assumption). Then g extends to $g' : v(B) \to I$, whence by adjunction a morphism $B \to u(I)$. Since $u(I)$ is injective, this morphism extends to $h : B' \to u(I)$, hence by adjunction a morphism $k : v(B') \to I$ extending g'. But then k "extends" g, which forces $A = 0$ since g was a monomorphism. Thus (a) is true. $\square$